Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.01 with the Adam optimizer. All experiments were run on a single GPU with 24 GB of memory.

2 Results

Figure 1 shows the accuracy by dataset.

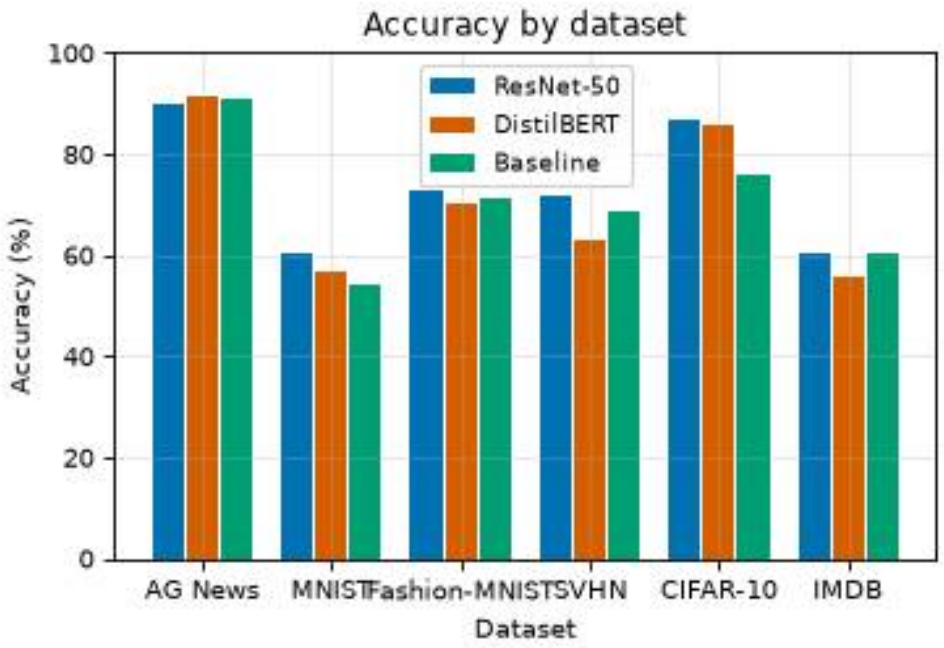


Figure 1: Accuracy by dataset for the models in this experiment.

Baseline outperforms DistilBERT by 6.9 percentage points on IMDB. On MNIST, DistilBERT obtains a accuracy of 76.6%.

Figure 2 shows the training loss.

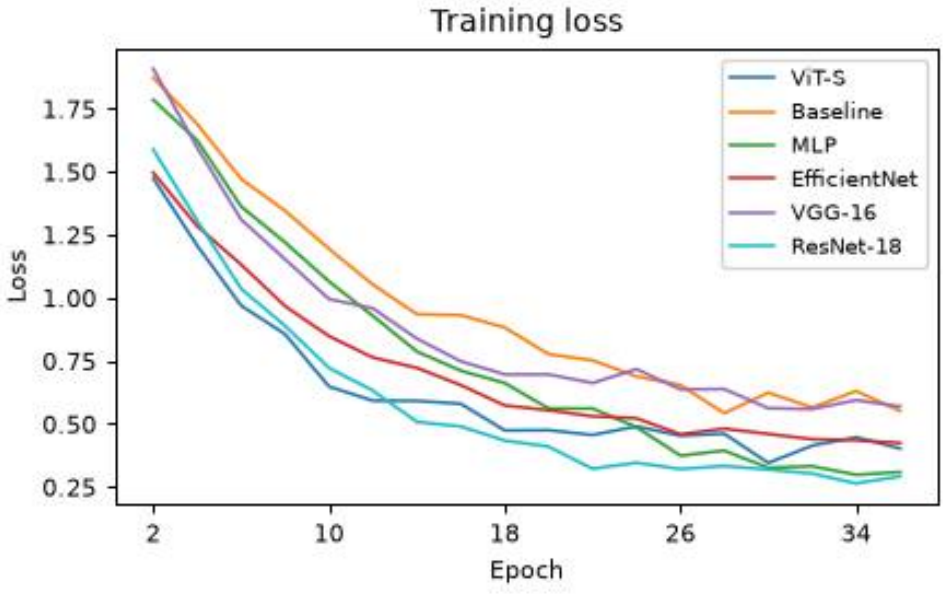


Figure 2: Training loss for the models in this experiment.

As shown in Figure 2, at epoch 30, the loss of EfficientNet is 0.279. VGG-16 has a loss of 0.838 at epoch 14. MLP has a loss of 1.213 at epoch 36. At epoch 36, the loss of ResNet-18 is 0.551.

Figure 3 shows the validation accuracy during training.

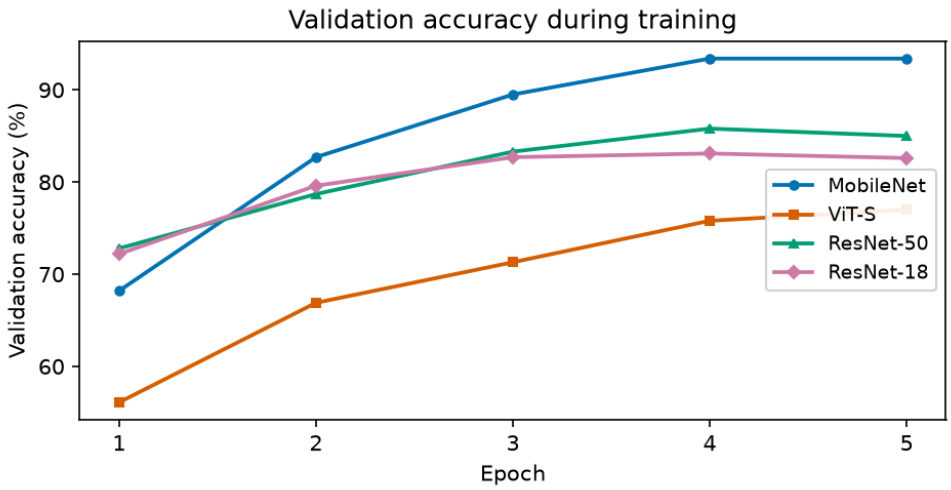


Figure 3: Validation accuracy during training for the models in this experiment.

MobileNet outperforms ViT-S by 20.3 percentage points at epoch 1 (Figure 3). As shown in Figure 3, at epoch 5, MobileNet achieves 94.2% validation accuracy. ResNet-50 reaches 75.1% validation accuracy at epoch 3. From epoch 1 to epoch 3, the validation accuracy of ResNet-18 improves by a relative about 10% (Figure 3).

3 Discussion

Naive Bayes reaches 61.0% accuracy on Tiny-ImageNet. We note that the training was stable for all models.